

Programme	BSc Mathematics Honours			
Course Code	MAT2VN101			
Course Title	LINEAR ALGEBRA FOR MACHINE LEARNING			
Type of Course	Vocational Minor – Introduction to AI			
Semester	II			
Academic Level	100-199			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	4	3	2	75
Pre-requisites	Foundations in Mathematics			
Course Summary	Course aims to provide basics of linear algebra which is useful in understanding machine learning problems			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Solve system of linear equations	Ap	C	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO2	Apply vector spaces and its properties	Ap	C	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO3	Understand basics of matrix algebra and its applications	U	C	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam

* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)
- Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)

Detailed Syllabus:

Textbook	Introduction to Linear Algebra" by Gilbert Strang, Wellesley-Cambridge Press, 2016, ISBN: 978-0980232776			
Module	Unit	Content	Hrs (45+ 30)	Marks (70)
I		Solving Linear Equations	12	Min.15
	1	Vectors and Linear Equation		
	2	The Idea of Elimination		
	3	Elimination Using Matrices		
	4	Rules for Matrix Operations		
	5	Inverse Matrices		
	6	Elimination = Factorization: $A = L U$		
	7	Transposes and Permutations		
II		Vector Spaces and Subspaces	12	Min.15
	8	Spaces of Vectors		
	9	The Nullspace of A: Solving $Ax = 0$		
	10	The Rank and the Row Reduced Form		
	11	The Complete Solution to $Ax = b$		
	12	Independence, Basis and Dimension		
	13	Dimensions of the Four Subspaces		
III		Orthogonality	8	Min.15
	14	Orthogonality of the Four Subspaces		
	15	Projections		
	16	Least Squares Approximations		
	17	Orthogonal Bases and Gram-Schmidt		
IV		Eigenvalues and Eigenvectors	13	Min.15
	18	Introduction to Eigenvalues		
	19	Diagonalizing a Matrix		
	20	Symmetric Matrices		

	21	Positive Definite Matrices		
	22	Similar Matrices		
	23	Singular Value Decomposition (SVD)		
		Practical using Python	30	
	1	Write Python function for vector operations: addition, scalar multiplication, norm,		
	2	Write Python function for matrix operations: addition, multiplication, inverse, transpose		
	3	Implement a Python function to solve a system of linear equations using NumPy's linear algebra module.		
	4	Implement matrix factorization techniques such as LU decomposition in Python using NumPy		
	5	Write a Python function to check if a set of vectors forms a vector space. And to determine if a set of vectors forms a subspace of a given vector space.		
	6	Write a Python function to find the basis of the column space, null space of a matrix, to calculate the rank, dimension of a matrix using NumPy,		
	7	Write a function to determine if a set of vectors is linearly independent, to find the span of a set of vectors. and to check if a set of vectors forms a basis for a given vector space.		
	8	Create a function to determine if two given vectors are orthogonal to each other and to calculate the projection of one vector onto another vector.		
	9	Use orthogonalization to find the least squares approximation of a vector that does not lie in the span of a given set of vectors.		
	10	Implement the Gram-Schmidt process in Python to orthogonalize a given set of vectors and to orthogonalize columns of a given matrix		
	11	Implement a function to perform a change of basis operation on a given vector.		
	12	Write a Python script to verify the rank-nullity theorem by computing the rank and nullity of a matrix and		

		comparing with the dimensions of its domain and codomain.		
	13	Write a Python function to compute the eigenvalues and eigenvectors of a square matrix using SciPy.		
	14	Write a Python function to check if a given square matrix is diagonalizable, to diagonalize a matrix using its eigenvectors and eigenvalues.		
	15	Write a Python function to compute the singular value decomposition of a matrix using NumPy, Use Singular Value Decomposition (SVD) to find the rank and dimension of a matrix, and discuss how it can be used for dimensionality reduction.		
		Reference		
	1	"Linear Algebra and Its Applications" by David C. Lay, Steven R. Lay, and Judi J. McDonald, Pearson, 2020, ISBN: 978-0134860244		
	2	Linear Algebra: Concepts and Applications" by Charles R. Johnson and Dean E. Riess, Wiley, 2017, ISBN: 978-1118612596		
	3	Linear Algebra: A Modern Introduction" by David Poole, Cengage Learning, 2016, ISBN: 978-1305658004		
	4	Linear Algebra for Machine Learning" by Jason Brownlee, Machine Learning Mastery, 2021		
	5	Numerical Python: Scientific Computing and Data Science Applications with Numpy, SciPy, and Matplotlib" by Robert Johansson, Apress, 2018, ISBN: 978-1484242452		

Note: Proofs of all the results are exempted for the end semester exam.

Mapping of COs with PSOs and POs :

	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	2	3	1	2	2	3	1	1
CO 2	3	2	3	1	2	2	3	1	1
CO 3	3	3	3	1	2	2	3	1	1

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓